

LW2 Tutorial: Sensitivity analysis and network models

Task 1 Sensitivity analysis

Ken and Larry, Inc., supplies its ice cream parlours with three flavours of ice cream: chocolate, vanilla, and banana. Due to extremely hot weather and a high demand for its products, the company has run short of its supply of ingredients: milk, sugar, and cream. Hence, they will not be able to fill all the orders received from their retail outlets, the ice cream parlours. Due to these circumstances, the company has decided to choose the amount of each flavour to produce that will maximize total profit, given the constraints on the supply of the basic ingredients.

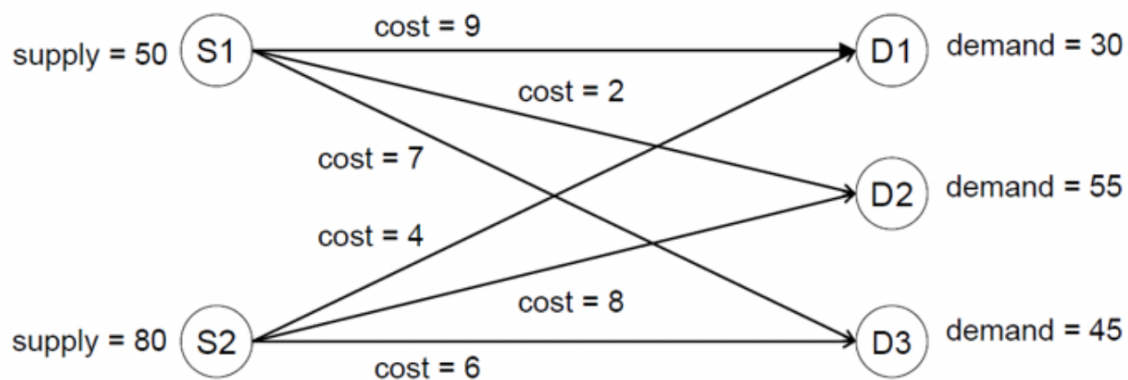
The chocolate, vanilla, and banana flavours generate, respectively, \$1.00, \$0.90, and \$0.95 of profit per gallon sold. The company has only 200 gallons of milk, 150 pounds of sugar, and 60 gallons of cream left in its inventory. The production of one gallon of each flavour requires the following amounts of ingredients:

	Chocolate	Vanilla	Banana
Milk (gallons)	0.45	0.50	0.40
Sugar (pounds)	0.50	0.40	0.40
Cream (gallons)	0.10	0.15	0.20

- (a) Formulate a linear programming model for this problem by clearly define the decision variables, objective function and constraints.
- (b) Build a spreadsheet model to identify the product mix which maximizes profit.
- (c) Produce a Sensitivity report and answer the following questions:
 - i. Suppose the profit per gallon of banana changes to \$1.00. Will the optimal solution change and what can be said about the effect on total profit?
 - ii. Suppose the profit per gallon of banana changes to 92¢. Will the optimal solution change and what can be said about the effect on total profit?
 - iii. Suppose the company discovers that three gallons of cream have gone sour and so must be thrown out. Will the shadow price change and what can be said about the effect on total profit?
 - iv. Suppose the company has the opportunity to buy an additional 15 pounds of sugar at a total cost of \$15. Should it do so? Explain.

Task 2 This task considers a typical supply-chain distribution problem faced by manufacturers and logistics planners. A firm operates two production plants with finite production capacities (shown in the Figure) and supplies three distribution centres (DCs) with known demand volumes (shown in the Figure). Transportation of product from plants to DCs incurs unit shipping costs that reflect distance, mode, handling complexity, and regional freight rates. The firm seeks an allocation of production shipments from plants to DCs that satisfies all DC demands while respecting plant capacities and minimising total transportation cost.

Model the problem with clearly defined decision variables, constraints and objective function and give suggestion about the distribution plan according to your solution.



Task 3 This task concerns an operational decision-making problem faced by community-based healthcare service providers. A regional healthcare centre manages a team of **10 visiting nurses** who are required to conduct **10 home visits** during a scheduled service period. Each home visit corresponds to a unique patient requiring clinical or supportive care, and each nurse is able to undertake exactly one assignment during the planning horizon.

The assignment of nurses to patients is associated with a quantifiable **cost**, provided in the accompanying Excel file. Each cost value represents a composite estimate derived from factors that commonly influence resource allocation in real-world community healthcare operations. Specifically, these factors include:

- **Travel distance and anticipated travel time** between the nurse's base location and the patient's residence.
- **Clinical task complexity**, recognising that different patients require varying levels of care, such as wound management, medication review, chronic disease monitoring, or mobility assistance.

These factors have been aggregated into a single cost metric for each potential nurse–patient pairing. Lower values indicate more favourable assignments from the

perspective of operational efficiency and clinical suitability. The healthcare centre seeks to determine the assignment configuration that minimises total system cost while ensuring that all patients are served.

You are required to:

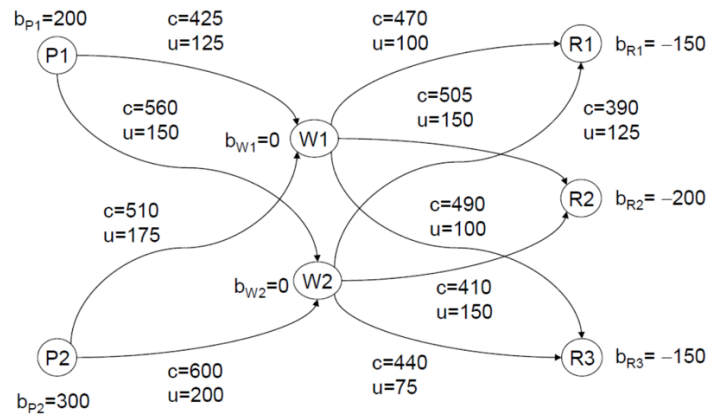
1. **Develop a complete algebraic formulation** of the problem, clearly defining all decision variables, parameters, constraints, and the objective function.
2. **Implement and solve** the formulated model using **Excel Solver**, enforcing binary decision variables and the necessary assignment constraints.
3. **Present the resulting optimal assignment** of nurses to patients, together with the corresponding minimum total cost, as produced by Solver.

Task 4 The Makonsel Company produces goods and sells them at its retail outlets. After production, the goods are stored in the company's two warehouses until needed by the retail outlets. Trucks are used to transport the goods from the two plants to the warehouses, and then from the warehouses to the three retail outlets. The following tables show the data for this scenario where the shipping capacity is given in units of full truckload. The problem is to find a distribution plan, i.e. number of full truckloads shipped per month from each plant to each warehouse and from each warehouse to each retail outlet, to minimise the total shipping cost.

To From	Unit Shipping Cost		Shipping Capacity		Output
	Warehouse 1	Warehouse 2	Warehouse 1	Warehouse 2	
Plant 1	£425	£560	125	150	200
Plant 2	£510	£600	175	200	300

To From	Unit Shipping Cost			Shipping Capacity		
	Retail Outlet 1	Retail Outlet 2	Retail Outlet 3	Retail Outlet 1	Retail Outlet 2	Retail Outlet 3
Warehouse 1	£470	£505	£490	100	150	100
Warehouse 2	£390	£410	£440	125	150	75
Demand	150	200	150	150	200	150

Graph depicting the company's distribution network.



- Formulate a linear programming model for this problem by clearly define the decision variables, objective function and constraints.
- Develop an Excel model and solve it by Excel Solver.

Optional Task: Work through the python file `transportation_probelm.ipynp` of the implementation of the transportation problem.